

Global Seismic Trends

Project Overview

The Global Seismic Trends project is designed to analyze earthquake data collected from around the world. The project uses earthquake information from the USGS Earthquake API and analyzes different factors such as earthquake magnitude, depth, location, tsunami occurrence, and data quality.

The main goal of this project is to understand earthquake patterns and identify important seismic trends using Python, Pandas, MySQL, SQL, and Streamlit.

The project also provides a dashboard where important findings can be viewed easily.

Project Objectives

The main objectives of this project are:

- Collect the last 5 years of earthquake data from the USGS Earthquake API.
- Clean and prepare the collected data for analysis.
- Identify countries/regions with high earthquake activity.
- Analyze earthquake magnitude and depth patterns.
- Study the relationship between earthquakes and tsunami events.
- Identify strong and destructive earthquakes.
- Store the cleaned data in MySQL.
- Perform SQL-based analysis on the earthquake dataset.
- Create a Streamlit dashboard to present the findings.

Data Source

The data used in this project is obtained from the USGS Earthquake API. This API provides information about earthquakes recorded around the world, including:

- Date and time of the earthquake
- Latitude and longitude
- Depth
- Magnitude
- Location
- Tsunami information
- Alert level
- Data quality information
- Earthquake identification details

Features in the Dataset

The dataset contains earthquake-related features as well as some derived features created during the data preparation stage.

Feature	Datatype	What it means	Why needed
id	str	ID is a unique identifier assigned to each earthquake event.	It can be used to identify a particular earthquake and avoid duplicating records.
time	datetime	The time feature represents the date and time when the earthquake occurred.	It is useful for studying earthquake activity over Year/Month/Day
updates	datetime	The updated feature represents the time when the earthquake information was last updated by USGS.	This can help understand when the earthquake record was modified or updated.
latitude	float	Latitude represents the north-south position of the earthquake.	It helps identify the geographical location of earthquake events and can also be used to study earthquake activity around the equator.
longitude	float	Longitude represents the east-west position of the earthquake.	Together, latitude and longitude help identify the location of an earthquake on the Earth's surface.
depth_km	float	Depth represents how far below Earth's surface the earthquake occurred.	Depth is useful for classifying earthquakes as shallow or deep.
mag	float	Magnitude represents the strength or size of an earthquake.	This feature is used in several analyses, including identifying the strongest earthquakes and calculating average magnitude by country or region.
magType	string	This indicates the method or scale used to calculate the earthquake magnitude.	Different magnitude calculation methods can be used by seismic monitoring system

place	string	The place feature contains the location description provided by USGS.	This contains information such as a nearby city, region, or distance from a particular location.
country	string	The country feature was derived from the earthquake location information.	It is useful for comparing earthquake activity between different countries.
status	string	Status represents the status of the earthquake information in the USGS system.	
tsunami	int	The tsunami feature indicates whether the earthquake was associated with a tsunami alert.	This feature is used to compare tsunami and non-tsunami earthquake events.
sig	int	The sig feature represents the significance score assigned to an earthquake event by USGS.	It can be used to identify earthquakes that have greater overall significance.
net	string	Identifies the network that reported the earthquake	Helps identify where the earthquake information came from
nst	float	nst represents the number of seismic stations that contributed information to the earthquake calculation.	It can be used to identify earthquakes that have greater overall significance.
dmin	float	dmin represents the distance to the nearest station used in the earthquake calculation.	It can be useful when studying the quality and reliability of earthquake measurements.
rms	float	RMS represents the root mean square error associated with the earthquake location solution	Lower RMS values generally indicate a better fit of the earthquake data.
gap	float	Gap represents the largest azimuthal gap between seismic stations used for determining the earthquake location.	It is one of the indicators that can be used to understand the quality of the earthquake location.

type	string	Tells what kind of event was recorded, such as earthquake, mining explosion, volcanic eruption, landslide, quarry blast, etc.	It is useful for event-type analysis and filtering
felt	float	Shows the number of people who reported the earthquake.	Helps understand the human impact of an earthquake.
cdi	float	The intensity of shaking was reported by people who experienced the earthquake.	Helps understand how strongly the earthquake was felt by people
mmi	float	An estimate of the intensity of shaking at a location	Helps measure the severity of shaking and its effects on people/buildings.
code	string	A code used by USGS to identify or reference the location/event associated with the earthquake.	Useful for identifying and grouping earthquake records
types	string	Lists the types of additional information/data available for an event	Helps to identify what supporting information is available for each earthquake event.
ids	string	Contains identifiers that link the event to USGS and other seismic networks/sources.	Useful for tracking the same event across different data sources
alert	string	The alert feature represents the alert level assigned to an earthquake.	The available alert levels include values such as: Green, Yellow, Orange, Red
year	int	The year was derived from the earthquake time.	It is used to analyze yearly earthquake trends and calculate year-over-year changes.
month	int	The month was derived from the earthquake date.	It helps identify monthly patterns in earthquake activity.
day	int	Day represents the day of the month when an earthquake occurred.	It helps to identify day-wise patterns in earthquake activity.
day_of_week	string	Day of Week was derived from the earthquake date.	It can be used to check whether earthquake activity differs across different days of the week.

depth_category	string	The earthquake depth was converted into categories to make analysis easier.	This classification helps compare shallow and deep earthquake activity.
magnitude_category	string	Magnitude was also converted into categories for easier analysis.	This helps in identifying earthquakes that require more attention because of their higher magnitude.

Data Collection

A single request for the complete five-year period resulted in more records than the API's maximum search limit. Therefore, the data was collected month by month.

The collected monthly data was then combined into a single DataFrame.

The data collection process included:

1. Setting the start and end dates.
2. Sending API requests for each month.
3. Checking whether the API request was successful.
4. Extracting earthquake properties.
5. Extracting latitude, longitude, and depth from the geometry information.
6. Converting timestamps into readable datetime values.
7. Combining all monthly data into one dataset.

Data Cleaning

After collecting the data, several data-cleaning steps were performed.

Handling Date and Time

The timestamp values provided by the API were converted into a datetime format. This made it easier to perform time-based analysis.

Handling Missing Values

Missing values were checked for all columns. Some columns such as nst, dmin, rms, gap, felt, cdi, mmi and alert contained missing values. Where appropriate, missing numerical values were handled using median values and 0.

Used median for nst, dmin, rms, and gap because these can have skewed distributions and outliers. For example, dmin goes up to 62.558, while most values are much smaller. nst also has a maximum of 619 while its median is only 32.

For felt, cdi, and mmi, missing values indicate that the corresponding impact information was not reported or available. Since these fields represent reported earthquake impact/intensity and zero is a valid value, I treated missing values as 0 to represent no reported value.

Duplicate Check

Duplicate records were checked to make sure that the dataset did not contain repeated earthquake events.

Creating Derived Features

Additional features were created from the existing data:

- Year
- Month
- Day
- Day of Week
- Country
- Depth Category
- Magnitude Category

These derived columns made the later SQL analysis and dashboard creation easier.

Database

After cleaning the data, the dataset was stored in a MySQL database using SQLAlchemy toolkit. SQLAlchemy used to connect Python with MySQL and load the cleaned dataset into the database.

The table name used in the project is: **earthquakes** and it contains the earthquake features and the derived columns required for analysis.

Data Analysis

Several SQL queries were created to understand global earthquake patterns.

The analysis includes:

Strongest Earthquakes

The earthquakes were sorted by magnitude to identify the strongest earthquake events. This helps identify the events with the highest magnitude during the last 5 year period.

Deepest Earthquakes

The data was sorted by earthquake depth to identify the deepest earthquake events. This helps understand where extremely deep earthquakes occurred.

Earthquake Activity by Country

The total number of earthquakes was calculated for each country. This helps identify countries that experienced the highest number of recorded earthquakes.

Average Magnitude by Country

The average earthquake magnitude was calculated for each country. This helps compare the general earthquake strength between different countries.

Shallow and Deep Earthquakes

Earthquakes were grouped based on their depth category. This analysis helps compare the number of shallow and deep earthquakes.

Earthquake Activity by Month

Earthquake events were grouped by month to identify monthly patterns. This helps understand whether some months recorded more earthquake activity than others.

Year-over-Year Earthquake Growth

The total number of earthquakes was calculated for each year. The year-over-year growth rate was then calculated to understand how earthquake activity changed from one year to the next.

Tsunami vs Non-Tsunami Earthquakes

Earthquakes were divided into two groups:

- Earthquakes associated with tsunami alerts
- Earthquakes without tsunami alerts

This comparison helps understand how frequently tsunami-related earthquake events occur.

Earthquakes Around the Equator

Countries with earthquakes occurring within a specified latitude range around the equator were analyzed. The average earthquake depth was calculated to understand the characteristics of earthquakes occurring near the equatorial region.

Shallow-to-Deep Earthquake Ratio

The ratio of shallow earthquakes to deep earthquakes was calculated for different countries. This helps identify countries where shallow earthquakes occur much more frequently than deep earthquakes.

Magnitude Difference Based on Tsunami Alerts

The average magnitude of earthquakes with tsunami alerts was compared with earthquakes without tsunami alerts. This helps understand whether tsunami-related earthquakes tend to have higher magnitudes.

Data Reliability Analysis

The gap and rms columns were used to study the reliability of earthquake location data. Higher error-related values can indicate less reliable earthquake measurements.

Earthquakes with High Station Coverage

The nst feature was used to identify earthquake events that had information from a larger number of seismic stations. This can help identify events for which more observation data was available.

Countries with Both Shallow and Deep Earthquakes

Countries were identified where both shallow and deep earthquakes occurred during the same month. This gives a better understanding of the variation in earthquake depth within the same geographical area.

Key Insights

The analysis provides several useful insights into global earthquake activity.

Earthquake frequency varies significantly by location

Some countries and regions experience considerably more recorded earthquakes than others. This is mainly related to their geographical position and tectonic activity.

Most earthquakes are not extremely high magnitude

Although the dataset contains some very strong earthquakes, most recorded events have comparatively lower magnitudes.

Shallow earthquakes are more common

The analysis shows that shallow earthquakes form a much larger portion of the dataset compared with deep earthquakes.

Tsunami-related earthquakes are a smaller group

Only a relatively small portion of the recorded earthquake events are associated with tsunami alerts.

Earthquake depth varies by region

Different countries and regions show different patterns in earthquake depth. Some areas experience both shallow and deep earthquakes.

Magnitude helps identify significant events

Magnitude is one of the most useful features for identifying strong earthquake events and comparing earthquake activity between regions.

Data quality is also important

Features such as nst, gap, dmin, and rms provide additional information about the quality of earthquake measurements.

Streamlit Dashboard

A Streamlit dashboard was created to present the analysis in a simple and interactive way. The dashboard provides an overview of the dataset and important earthquake statistics.

The main summary metrics include:

- Total Earthquakes
- Average Magnitude
- Maximum Magnitude
- Tsunami Events
- Deep Earthquakes

The dashboard also includes visualizations for different earthquake trends.

Examples include:

- Earthquake activity by year
- Tsunami vs non-tsunami earthquakes
- Top 5 Earthquake activity by country
- Magnitude Over Time
- Other important analysis results

The purpose of the dashboard is to make the analysis easier to understand without requiring the user to read SQL queries or Python code.

Technologies Used

The following technologies were used in the project:

Technology	Usage
Python	Used for API data collection, data cleaning, transformation, and analysis.
Pandas	Used for loading, cleaning, and transforming dataset.
SQLAlchemy	Used to connect Python with the MySQL database.
SQL	Used to store the cleaned earthquake dataset.
Streamlit	Used to create an interactive dashboard.

Future Analysis and Enhancements

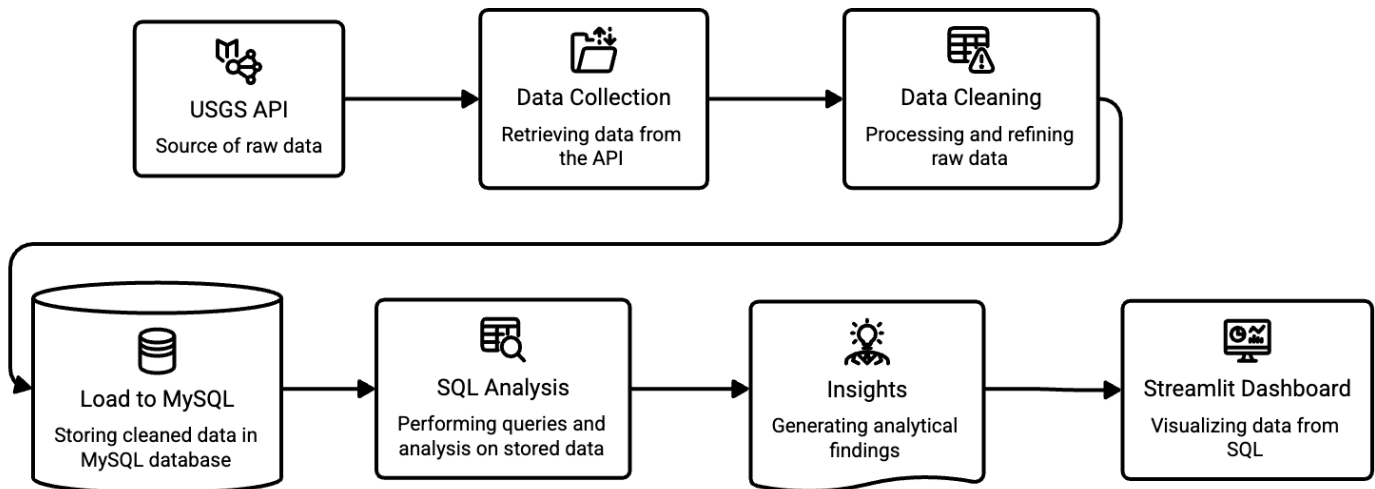
The current project provides useful insights into global earthquake activity using historical USGS data. The project can be further improved with the following enhancements:

- **Real-Time Monitoring and Automatic Updates:** Connect the dashboard to the USGS API and automatically fetch new earthquake data at regular intervals, allowing the dashboard to display the latest seismic activity without manual updates.
- **Interactive Geographic Analysis:** Add an interactive world map that allows users to explore earthquake activity by country or region.
- **Machine Learning Analysis:** Apply machine learning techniques to explore patterns in earthquake magnitude, depth, and other earthquake characteristics.
- **Earthquake Impact Analysis:** Use information such as felt reports, CDI, MMI, and tsunami events to study the potential impact of significant earthquakes.

These enhancements can make the project more interactive and help provide deeper insights into global seismic activity.

Project Workflow

The overall workflow of the project is:



Conclusion

The Global Seismic Trends project provides a detailed analysis of earthquake activity across different countries and regions. The project combines API data collection, data cleaning, SQL analysis, and visualization to understand earthquake patterns.

By analyzing features such as magnitude, depth, location, tsunami information, and data quality, the project provides useful insights into global seismic activity. The Streamlit dashboard makes these insights easier to understand through simple metrics and visualizations.

Overall, this project demonstrates how real-world data can be collected, cleaned, stored, analyzed, and presented using Python, SQL, and data visualization tools.